MILITARY SPECIFICATION

INCREMENTS, PROPELLANT FOR CHARGE, PROPELLING, M36A2 (MADE FROM PROPELLANT, M8)

1. SCOPE

1.1 This specification covers propellant increments made from M8 Propellant for use in Charge, Propelling, M36A2 for 4.2 Inch Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Strapping, Flat; Steel

MILITARY

MIL-B-2427	-	Box, Ammunition Packing; Wood, Nailed
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for
MIL-B-3106	-	Board, Composition, Water-Resistant, Solid
MIL-L-10547	-	Liners, Case, Waterproof, Flexible
MIL-B-13239	-	Barrier Material, Waterproofed, Flexible All Temperature
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance

FSC: 1315

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-414	-	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-652	-	Propellants, Solid, for Cannons; Requirements and Packing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8796522	-	Marking Diagram and Sealing for Wooden Packing Boxes
9241693	-	Packing and Marking of Fiberboard Cartons for Shipment of Propellant Increments
76-4-46	-	Box, Packing, with Metal Liner, M24 for Smokeless Powder; Assembly and Details
76-4-56	-	Box, Packing, Metal-Wood, M17 for Smokeless Powder
76-4-59	-	Box, Packing, Metal-Wood, M18 for Smokeless Powder Assembly
7548645	-	Carton, Packing, Reusable Collapsible for High Explosive, Assembly Details, Packing and Marking
7549033	-	Container, Metal, Universal, M25 for Propellants and Explosives
8858577	-	Marking Diagram and Sealing of Container, Metal, Universal M25 for Shipment of Propellants
8858848	-	Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellants
9244177	-	Charge, Propelling: M36A2
9244179	-	Half Increment
9244180	-	Increment
9244178	-	Five Increment Bundle

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9276822 - Propelling Charge Sub Assembly (Front)
9276823 - Propelling Charge Sub Assembly (Rear)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 171 - 179

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402. Orders for the above publication should cite: 49 CFR 171-179 (latest revision).

3. REQUIREMENTS

3.1 Material. - The propellant increments shall be made from M8 Propellant which shall conform to the requirements of MIL-STD-652 and the applicable drawings.

3.2 Half Increment, Increment and Five Increment Bundle. - The Half Increment, Increment and Five Increment Bundle shall comply with all requirements specified on Drawings 9244179, 9244180 and 9244178 respectively and with all requirements specified in this specification and in all other applicable specifications.

3.3 Ballistic Acceptance Test. - The propellant increments shall comply with the following requirements, when assembled to 4.2 inch test rounds and tested at the 34 increment level (maximum charge). Testing shall be in accordance with paragraph 4.3 using samples chosen in accordance with paragraph 4.2.3.2.

3.3.1 Muzzle Velocity. - The corrected muzzle velocity shall not be more than 1040 feet per second (fps) nor less than 1010 fps.

3.3.2 Velocity Standard Deviation. - Velocity standard deviation shall not exceed 5.0 fps.

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3.3.3 Pressure (Average). - The average pressure shall be no greater than 15,600 pounds per square inch (psi).

3.3.4 Pressure (Maximum Permissible Individual). - The Maximum Permissible Individual Pressure of any sample shall be no greater than 17,900 psi. (115 percent of the upper pressure limit of the weapon).

3.4 Workmanship. - The propellant, and the ingredients which compose the propellant, shall be protected from the action of direct sunlight and acid fumes. The propellant increments shall show no visible evidence of moisture and shall be free of cracked, torn, deformed or otherwise distorted propellant sheets. The stitches, bags and seals shall be without breaks, tears, grease or noticeable foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109, in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of batches of product inspected (shall be made available upon request by the contracting officer).
c. Results obtained for all inspections performed (shall be made available upon request by the contracting officer).
d. Drawing, specification number, and date, together with an identification and date of changes.

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e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contracting documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot.

g. Date submitted.

h. Eight copies of propellant description sheet SMU Form 1060 containing the applicable data. (see 6.5)

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. - The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots or one or more batches of M8 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M8 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Half Increment (see dwg. 9244179)

Categories	Defects	Method of Inspection
Critical:		
	1. Incorrect Weight	Scale CD
Major:		
	101. Diameter of hole	Gage CD
	102. Length of sides	Gage CD
	103. Thickness	Gage CD
	104. Evidence of poor workmanship (see 3.4)	Visual
Minor: None Defined		

4.2.2.2 Increment (see dwg. 9244180)

Categories	Defects	Method of Inspection
Critical:		
	1. Incorrect weight	Scale CD
Major:		
	101. Evidence of poor workmanship(see 3.4)	Visual
Minor:		
	201. Radial cuts not in same plane	Visual

4.2.2.3 Five Increment Bundle (see dwg. 9244178).

Categories	Defects	Method of Inspection
Critical:		
	1. Incorrect weight	Scale CD
Major:		
	101. Evidence of poor workmanship(see 3.4)	Visual
Minor:		
	201. Radial cuts not in same plane	Visual

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4.2.2.4 Propelling Charge Sub-Assembly (Front) see dwg. 9276822.

Categories	Defects	Method of Inspection	
Critical:			
1.	Incorrect weight (weighing to be done immediately prior to assembling increments to shipping spindle).....	Scale	CD
Major:			
101.	Evidence of poor workmanship(see 3.4)...	Visual	
Minor: None defined			

4.2.2.5 Propelling Charge Sub-Assembly (Rear) see dwg. 9276823.

Categories	Defects	Method of Inspection	
Critical:			
1.	Incorrect weight (weighing to be done immediately prior to assembling increments to shipping spindle.....	Scale	CD
Major:			
101.	Evidence of poor workmanship(see 3.4)...	Visual	
Minor: None defined			

4.2.2.5 Sealed inner liner (see drawing 7548645).

Categories	Defects	Method of Inspection
Critical: None defined		
Major:		
101.	Liner damaged so that contents are exposed or liable to become exposed.....	Visual
102.	Bag incompletely sealed.....	Visual
Minor: None defined		

4.2.2.6 Sealed inner liner (see drawing 7548645).

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Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Liner damaged so that contents are exposed or liable to become exposed....	Visual
102.	Bag incompletely sealed.....	Visual
Minor:	None defined	

4.2.2.7 Sealed wood packing box (see paragraph 5.1.1 if applicable). Sealed fiber board carton (see drawing 9241693 if applicable).

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Banding or strapping missing, broken or loose.....	Visual-Manual
102.	Board broken or split.....	Visual
103.	Contents exposed.....	Visual
Minor:		
201.	Banding or strapping improperly engaged.....	Manual
202.	Contents move when shaken.....	Manual
203.	Marking missing, misleading, or illegible.....	Visual
204.	Nails, screws or fasteners loose.....	Visual

4.2.2.8 Box or container, prior to filling (see drawing 76-4-46, 76-4-59, 7549033, or 76-4-56, as applicable).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Cracks or holes in metal liner (as applicable).....	Visual
102.	Foreign matter on interior.....	Visual
Minor:	None defined	

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4.2.2.9 Sealed box or container (see drawing 76-4-46, 76-4-59, or 76-4-56, 7549033, 8858848 or 8858577, as applicable).

Categories	Defects	Method of Inspection
Critical:	None defined	
Major:		
101.	Cover incompletely closed	Visual-Manual
102.	Gasket missing or broken	Visual
103.	Wooden cover split or otherwise damaged (when applicable)	Visual
Minor:		
201.	Car seal missing, unsealed or improperly positioned	Visual-Manual
202.	Crack or hole in metal container	Visual
203.	Hardware improperly engaged	Visual
204.	Marking misleading or unidentifiable	Visual
205.	Bare areas in exterior coating of metal container the sum or which is in excess of 1/4 square inch	Visual

4.2.3 Testing

4.2.3.1 Material (see 3.1).-Each lot of propellant shall be sampled and tested in accordance with MIL-STD-652.

4.2.3.2 Ballistic Acceptance Test (see 3.3).-Twenty-five complete sets of M36A2 Propelling Charges (see dwg. 9244177) shall be randomly selected from each Propellant Increment lot and fired for acceptance at a proving ground. Observation shall be made for compliance with ballistic acceptance requirements as follows:

4.2.3.2.1 Muzzle Velocity.-Major Defect (see 3.3.1).-The sampling plan and acceptance criteria shall be in accordance with MIL-STD-414, Section B Code Letter I, and an AQL of 2.5 percent.

4.2.3.2.2 Velocity Standard Deviation.-Major Defect.-The propellant increment lot shall be rejected if the velocity standard deviation multiplied by the factor 0.81 (see 6.4) fails to comply with the requirement in para. 3.3.2.

4.2.3.2.3 Pressure (Average).-Critical Defect.-Failure of the average pressure to comply with the requirement, as stated in para. 3.3.3, shall be cause for propellant increment lot rejection.

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4.2.3.2.4 Pressure (Maximum Permissible Individual).- If any individual pressure of any sample exceeds the requirement, as stated in para. 3.3.4, the propellant increment lot shall be rejected.

4.2.4 The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 (see 6.2).

4.2.4.1 Inspection Equipment Capability and Accuracy.-All inspection equipment and facilities shall be of adequate quality and quantity to perform properly and accurately all the required inspections, measurements and tests. The inspection equipment shall be maintained in satisfactory condition for the continued proper performance of these functions and shall be calibrated in accordance with the provisions of MIL-C-45662.

4.3 Test Methods and Procedures

4.3.1 Proving Ground Tests.-These tests shall be performed at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure.

4.3.2 Ballistic Acceptance Test.-The M329A2, 4.2" test cartridges (see dwg. 9235654) shall be assembled with M36A2 Propelling Charges (34 increments with bag) chosen in accordance with para. 4.2.3.2. The rounds shall be inert filled to 21.99 ± .12 pounds. This weight shall be without the M36A2 Propelling Charge, Propellant Holder and Obturator. Ten (10) reference rounds shall be similarly assembled. However, reference M36A2 Propelling Charges shall be used. Both test and reference rounds shall be conditioned and fired at 70 degrees F. Test round pressure and test round velocity data shall be corrected using the reference round data to eliminate firing condition error. Temperature of the day, weather and time of firing shall be recorded. The velocity average and velocity standard deviation shall be calculated after correcting results to standard conditions, and all firing data (includes individual velocity and pressure data) shall be reported to Picatinny Arsenal for evaluation (see 6.2).

5. PREPARATION FOR DELIVERY

5.1 Packing (see 6.1).

5.1.1 Level A.-Propellant increments shall be packed im case liners conforming to Type I, Grade A, Style A, Specification MIL-L-10547; manufactured from barrier material conforming to

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Class 1, Specification MIL-B-131. The case liner shall be approximately 14 3/4 inches long by 14 3/4 inches wide by 18 1/2 inches deep. The case liner shall be closed with pressure sensitive tape, Type, Type IV, Class 1 of Specification PPP-T-60. One liner shall be packed in a nailed wood box complying with Grade A, Type I, Class 1 of Specification MIL-B-2427. The box shall be completely lined with fillers complying with specification MIL-F-50449, The box shall be strapped girthwise, between cover cleats with two 5/8 inch by 0.02 inch thick flat steel straps conforming to Type 1, Class A or B, Specification QQ-S-781. The net weight of the propellant shall not exceed 50 pounds.

5.1.2 Level B.-Propellant increments shall be packed in accordance with paragraph 4.3 of MIL-STD-652.

5.1.3 Level C.-Propellant increments shall be packed in a fiberboard carton conforming to requirements for Carton No. 1 of Drawing 7548645 except that liner shall conform to Type 1, Grade A material of Specification MIL-L-10547. Alternatively, a wood box meeting the requirements of Code of Federal Regulations Parts 0-190 may be used in lieu of fiberboard carton.

5.2 Marking

5.2.1 Level A.-Exterior containers shall be marked in accordance with drawing 8796522. The DOT marking shall be "Propellant Explosive - Class A."

5.2.2 Level B.-The M17, M18, M24 containers shall be marked in accordance with drawing 8858848. The M25 container shall be marked in accordance with drawing 8858577.

5.2.3 Level C.-The fiberboard carton shall be marked in accordance with drawing 9241693.

5.3 Transportation.-Transportation of propellant increments shall be in accordance with publication listed in para. 2.2.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1).
c. Process.-Details of the manufacturing process and the equipment used by the contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Submission of test data.-In addition to the normal distribution of records, one copy of all ballistic test data shall be forwarded to: Commanding Officer, Picatinny Arsenal,

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ATTN: SMUPA-QA-A-A, Dover, New Jersey 07801 and one copy to ATTN: SMUPA-ADEP-2.

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified any may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or

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other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Factor 0.81.-The factor 0.81 makes allowance for the probability that a standard deviation from a twenty five round sample exceeds the true standard deviation by chance alone.

6.5 Submission of SMU Form 1060.-Information contained in propellant description sheet, SMU Form 1060, need not be resubmitted in the form of a certificate provided that the provisions regarding the certifying or responsible agents are adhered to (see 4.1.1).

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1315-A-021

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